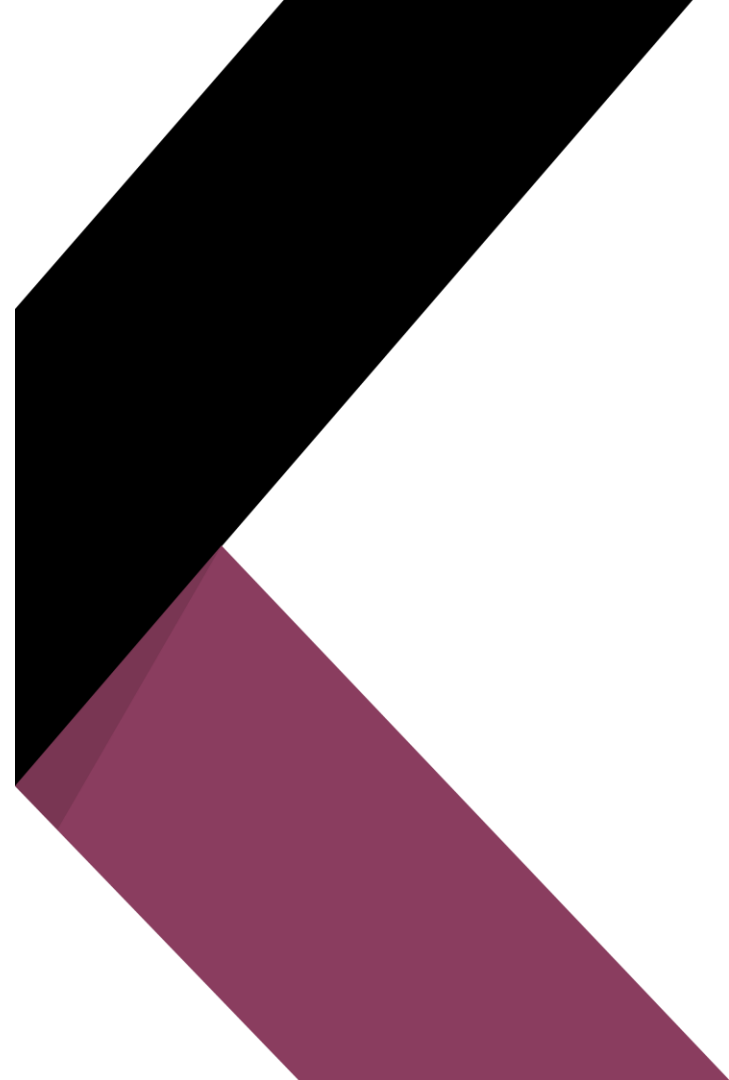


Requirements' management





Requirements' management - Outline

- ❑ Automotive SPICE
- ❑ A good requirement
- ❑ Rational DOORS



Automotive SPICE



ASPICE - Introduction

Software Process Improvement and Capability Determination (Evaluation)

- Automotive SPICE is a domain-specific variant of the international standard ISO / IEC 15504 (SPICE).
- The purpose of **Automotive** SPICE is to improve and evaluate the performance of the development processes of ECU suppliers in the **automotive** industry.





ASPICE - Overview

Requirement Elicitation

Gather, process, and track evolving stakeholder needs and requirements throughout the lifecycle of the product

System Requirement Analysis

Transform the defined stakeholder requirements into a set of system requirements that will guide the design of the system

Software Requirement Analysis

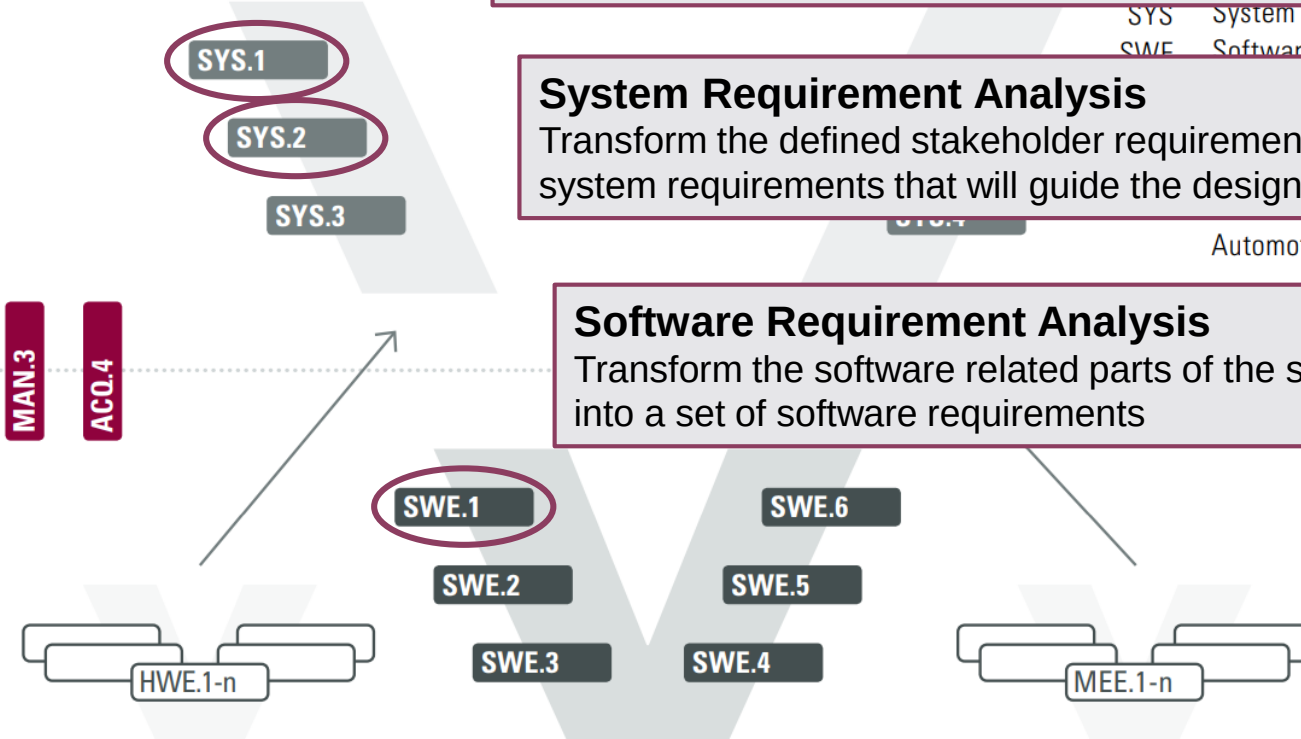
Transform the software related parts of the system requirements into a set of software requirements

SYS System Engineering
SWE Software Engineering

Automotive SPICE® v3.1

System level

Domain level





A good requirement



Requirement - Characteristics

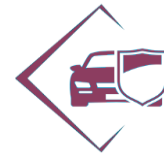
Functional Requirement

What a system shall *do*.



Non-functional Requirement

What a system shall *be*.





Requirement - Characteristics

Individual Requirement

- ☐ Clear and Unambiguous
- ☐ Testable (verifiable)
- ☐ Feasible
- ☐ Consistent
- ☐ Atomic
- ☐ Traceable

There should be only **1** way to interpret the requirement.

✗ **REQ:** *The system shall not accept password longer then 15 characters.*

✓ **REQ:** *If the user inserts a password longer then 15 characters, then the system shall display an error message and shall ask the user to correct it.*



Requirement - Characteristics

Individual Requirement

- ☐ Clear and Unambiguous
- ☐ Testable (verifiable)
- ☐ Feasible
- ☐ Consistent
- ☐ Atomic
- ☐ Traceable

The requirement can be verified by means of: inspection, analysis, demonstration or test.

✗ **REQ:** *The system must refresh the data reasonably quickly.*

✓ **REQ:** *The system must refresh the data each 0.5 ms.*



Requirement - Characteristics

Individual Requirement

- ☐ Clear and Unambiguous
- ☐ Testable (verifiable)
- ☐ Feasible
- ☐ Consistent
- ☐ Atomic
- ☐ Traceable

The requirement should be doable within existing constraints.

✗ **REQ:** *The replacement control system shall be installed with no disruption of the production.*

✓ **REQ:** *The replacement control system shall be installed causing no more then 2 days of production disruption.*



Requirement - Characteristics

Individual Requirement

- ☐ Clear and Unambiguous
- ☐ Testable (verifiable)
- ☐ Feasible
- ☐ Consistent
- ☐ Atomic
- ☐ Traceable

There should not be any conflicts between the requirements.

- ✗ **REQ1:** *If $A \neq B$, then initialization shall be triggered.*
- ✗ **REQ2:** *The request for initialization shall be sent permanently with the value 0 ('kein_init').*

Avoid redundancy!

- ✓ **REQ:** *The request for initialization shall be sent permanently with the value 0 ('kein_init').*



Requirement - Characteristics

Individual Requirement

- ☐ Clear and Unambiguous
- ☐ Testable (verifiable)
- ☐ Feasible
- ☐ Consistent
- ☐ Atomic
- ☐ Traceable

Requirements should be “singular” and include only one requirement with no use of conjunctions.

✗ **REQ:** *In case of overtemperature, overcurrent and overvoltage the system shall abort the charge.*

- ✓ **REQ1:** *In case of overtemperature the system shall abort the charge.*
- ✓ **REQ2:** *In case of overcurrent the system shall abort the charge.*
- ✓ **REQ3:** *In case of overvoltage the system shall abort the charge.*



Requirement - Characteristics

Individual Requirement

- ☐ Clear and Unambiguous
- ☐ Testable (verifiable)
- ☐ Feasible
- ☐ Consistent
- ☐ Atomic
- ☐ Traceable



It must be always possible to know to which level a requirement belongs to and its correlation with other requirements.



Rational DOORS





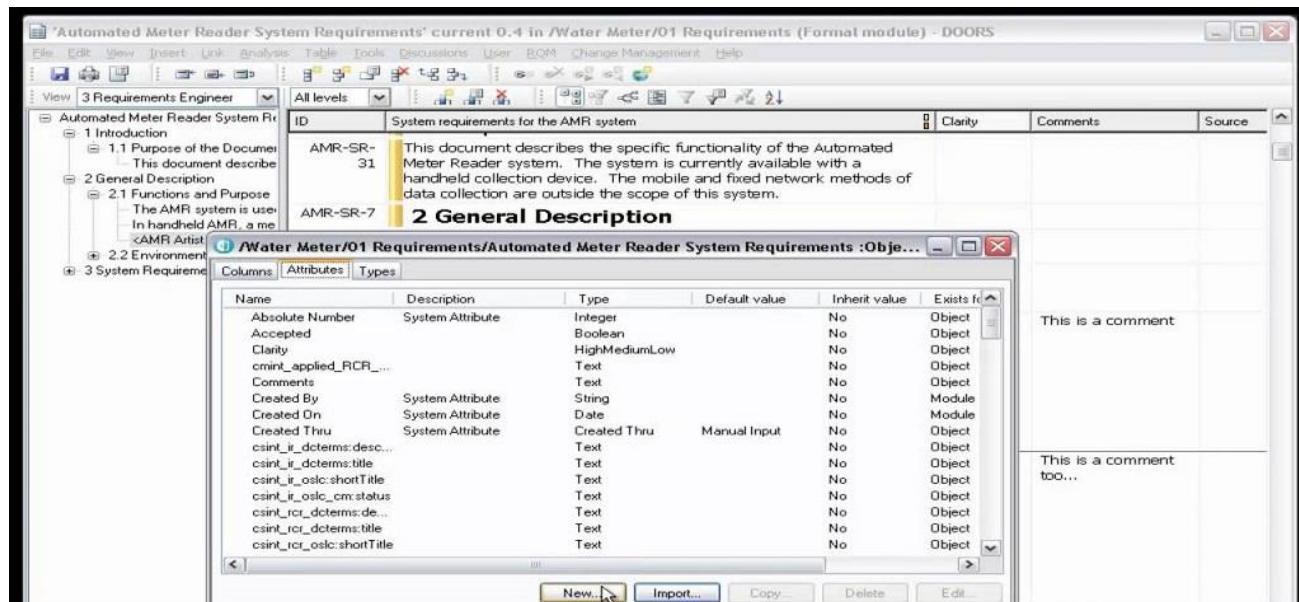
DOORS - Attributes

```
Module m = current
Object o = current
Filter f

load view "Standard view"
```

DOORS output

Run Next error Print Load Save As... Browse Close Help



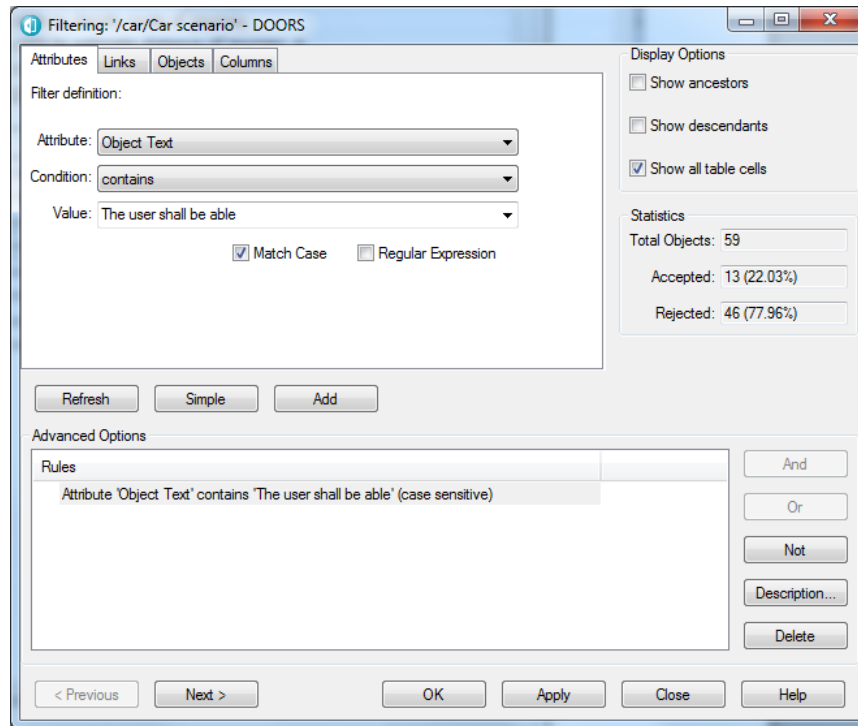
- Characterize the requirement object.
- Supports the process.
- Allows the efficient and rational administration of requirements.



DOORS -

Attributes

- ✓ Filtering through specific:
 - Names
 - IDs
 - Links through modules
 - Wildcards
 - Predefined attributes
 - Customized attributes
- ✓ Multiple conditions applicable.
- ✓ Regular expression.
- ✗ No long memory!





DOORS -

Attributes

➤ Baselines

➤ Links for traceability

➤ Defined views for exports

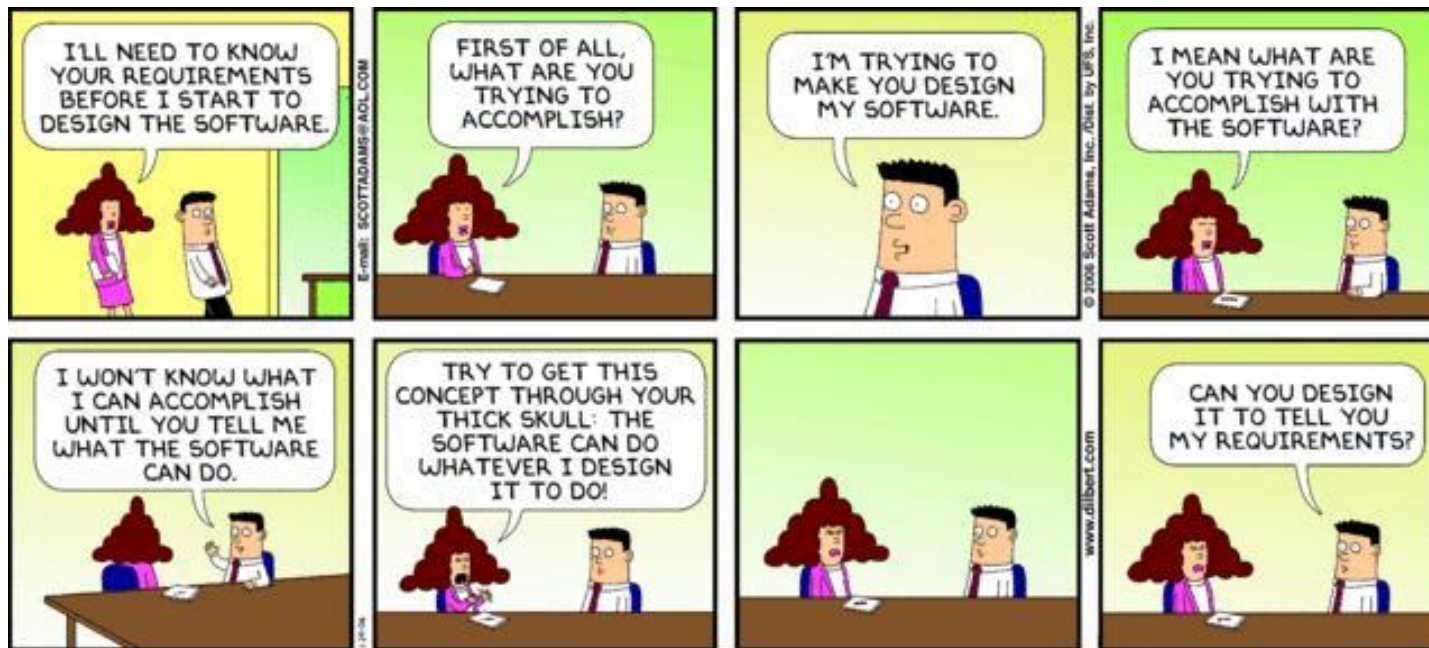
➤ Easy import/update attributes

➤ History



Conclusion

The end..?



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